

DevAlpha Technologies
Civil Engineering & Structural Design Internship

TASK 3 — STRUCTURAL ANALYSIS PROJECT

Study of Load Analysis & Safety Factors in Structural Engineering

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1. Executive Summary

This Structural Analysis Project investigates the fundamental principles governing load behavior and safety factor application in civil engineering structures. The report covers identification and classification of structural loads, applicable load combination methods per international design standards, and the derivation of safety factors used to ensure structural integrity under worst-case scenarios.

A representative five-story reinforced concrete (RC) office building is used as the primary case study to demonstrate practical load calculation, structural verification, and failure mode identification. The analysis follows the guidelines set by ACI 318-19, ASCE 7-22, and IS 456:2000.

Key outcomes of this report include computed dead, live, wind, and seismic loads; load combination results; factor of safety (FOS) checks; and design recommendations to ensure code compliance and occupant safety.

2. Introduction

2.1 Background

Structural analysis is the backbone of safe construction. Every structure — from residential homes to high-rise towers — must resist a multitude of forces without experiencing excessive deformation or catastrophic failure. Understanding the nature of loads and applying appropriate safety margins is a core competency for any structural engineer.

2.2 Objectives

- Classify and quantify the various load types acting on a typical structure.
- Apply standard load combination methods as per ASCE 7-22 / IS 456:2000.
- Determine and validate safety factors for structural members.
- Perform a hands-on load analysis using a five-story RC office building case study.
- Identify potential failure modes and propose design improvements.

2.3 Scope

This report is limited to gravitational, wind, and seismic load analysis of a reinforced concrete framed structure. Soil-structure interaction, foundation design, and dynamic time-history analysis are outside the scope of this project.

3. Types of Structural Loads

3.1 Dead Loads (DL)

Dead loads are permanent, static forces arising from the self-weight of structural and non-structural elements. They are constant throughout the life of the structure.

- Self-weight of slabs, beams, columns, and walls.
- Weight of permanent fixtures: flooring, ceiling finishes, HVAC systems.
- Typical slab self-weight: 3.75 kN/m^2 (150 mm RC slab at 25 kN/m^3).
- Superimposed dead load (SDL): $1.5\text{--}2.0 \text{ kN/m}^2$ for finishes.

3.2 Live Loads (LL)

Live loads are variable, transient forces due to occupancy, furniture, and movable equipment. They depend on building use and occupancy category.

- Office areas: 2.5 kN/m^2 (ASCE 7-22, Table 4.3-1).
- Corridors and lobbies: 4.8 kN/m^2 .
- Roof live load: 1.0 kN/m^2 (accessible roof).
- Stairways: 4.8 kN/m^2 .

3.3 Wind Loads (WL)

Wind loads are lateral forces caused by wind pressure acting on the exterior surfaces of a structure. They are calculated based on geographic location, terrain exposure, and building height.

- Design wind speed (V_b): 44 m/s (Terrain Category II, Risk Category II).
- Wind pressure formula: $q = 0.5 \times \rho \times V_d^2$ where $\rho = 1.2 \text{ kg/m}^3$.
- Gust factor (G): 0.85 for rigid structures.
- Pressure coefficient (C_p): 0.8 (windward), -0.5 (leeward).

3.4 Seismic / Earthquake Loads (EL)

Seismic loads are inertial forces induced by ground motion during earthquakes. They are a function of the building's weight, stiffness, and seismic zone.

- Seismic Zone: Zone III (Moderate — IS 1893:2016).
- Zone factor (Z): 0.16.
- Response reduction factor (R): 5 (SMRF).
- Importance factor (I): 1.0.
- Base shear formula: $V = (A_h \times W)$ where $A_h = (Z/2) \times (S_a/g) / (R/I)$.

3.5 Other Load Types

Load Type	Description	Typical Application
Snow Load	Accumulation of snow on roofs	Cold climates, IS 875 Part 4
Temperature Load	Thermal expansion/contraction	Bridges, long buildings
Hydrostatic Load	Water pressure on retaining structures	Basements, dams, tanks
Impact Load	Sudden dynamic forces	Industrial facilities, parking garages
Construction Load	Temporary loads during building	Formwork, scaffolding

4. Load Combination Methods

4.1 ASCE 7-22 LRFD Combinations

Load and Resistance Factor Design (LRFD) applies partial safety factors to each load type to obtain factored design loads. The governing combination produces the maximum demand on a structural member.

#	Load Combination	Notes
1	1.4(DL)	Gravity only — no live/lateral load
2	1.2(DL) + 1.6(LL) + 0.5(Lr)	Dominant gravity combination
3	1.2(DL) + 1.6(Lr) + (LL or 0.5WL)	Roof live + wind interaction
4	1.2(DL) + 1.0WL + LL + 0.5(Lr)	Wind governs lateral
5	0.9(DL) + 1.0WL	Uplift check (overturning)
6	1.2(DL) + 1.0EL + LL	Seismic combination
7	0.9(DL) + 1.0EL	Seismic uplift check

4.2 IS 456:2000 / IS 875 Combinations (Indian Standard)

- $1.5(DL + LL)$ — Basic gravity combination.
- $1.5(DL + EL)$ or $1.5(DL + WL)$ — Lateral load dominant.
- $1.2(DL + LL + WL)$ or $1.2(DL + LL + EL)$ — Combined effect.
- $0.9(DL) \pm 1.5(EL)$ — For uplift and overturning.

4.3 Critical Combination Selection

For the case study RC office building, Combination 2 [$1.2DL + 1.6LL + 0.5Lr$] governs floor beam design, while Combination 6 [$1.2DL + 1.0EL + LL$] governs column and shear wall design due to the moderate seismic zone classification.

5. Safety Factors & Design Codes

5.1 What Is a Factor of Safety (FOS)?

The Factor of Safety is the ratio of the structural capacity (resistance) to the applied demand (load). It accounts for uncertainties in material properties, load estimation, construction quality, and analytical assumptions.

$$\text{FOS} = \text{Structural Resistance (Rn)} / \text{Applied Load Effect (Sn)} \geq 1.0$$

5.2 Material Safety Factors (γ_m)

Material	Code Standard	γ_m (Basic)	γ_m (Seismic)
Concrete (compression)	IS 456 / ACI 318	1.50	1.50
Reinforcement Steel	IS 456 / ACI 318	1.15	1.00
Structural Steel	IS 800 / AISC 360	1.10	1.00
Masonry	IS 1905	1.50	—
Timber	IS 883	2.50	—

5.3 Load Partial Safety Factors (γf)

Load Type	Comb. 1 (Grav.)	Comb. 2 (Wind)	Comb. 3 (Seism.)	Uplift Check
Dead Load (DL)	1.5	1.2	1.2	0.9
Live Load (LL)	1.5	1.2	1.0	—
Wind Load (WL)	—	1.5	—	1.5
Seismic Load (EL)	—	—	1.5	1.5

5.4 Minimum Required FOS by Element Type

- Columns (axial compression): $FOS \geq 3.0$
- Beams (bending/shear): $FOS \geq 1.7 - 2.0$
- Footings (bearing capacity): $FOS \geq 2.5 - 3.0$
- Retaining Walls (overturning): $FOS \geq 2.0$
- Retaining Walls (sliding): $FOS \geq 1.5$
- Slopes and embankments: $FOS \geq 1.5$

6. Structural Analysis — Case Study

6.1 Building Description

Building Type	Reinforced Concrete (RC) Framed Office Building
Number of Stories	5 (Ground + 4 Upper Floors)
Floor Height	3.5 m per story
Total Building Height	17.5 m
Plan Dimensions	20 m × 15 m (L × B)
Floor Area per Story	300 m²
Total Built-up Area	1,500 m²
Slab Thickness	150 mm two-way RC slab
Column Grid	5 m × 5 m (16 columns total)
Structural System	Moment Resisting Frame (SMRF)
Concrete Grade	M30 (fck = 30 N/mm²)
Steel Grade	Fe500 (fy = 500 N/mm²)
Seismic Zone	Zone III — IS 1893:2016

Soil Type	Medium Soil (Type II)
Wind Zone	Zone IV — IS 875 Part 3

6.2 Dead Load Calculation

Floor Dead Loads:

- RC Slab self-weight: $0.15\text{ m} \times 25\text{ kN/m}^3 = 3.75\text{ kN/m}^2$
- Floor finishes (tiles + screed): 1.00 kN/m^2
- Ceiling & services (MEP): 0.50 kN/m^2
- Partition walls (movable): 1.00 kN/m^2

Total Floor Dead Load (DL) = $3.75 + 1.00 + 0.50 + 1.00 = 6.25\text{ kN/m}^2$

Roof Dead Loads:

- RC Slab self-weight: 3.75 kN/m^2
- Waterproofing layer: 0.50 kN/m^2
- Roof finishes: 0.75 kN/m^2

Total Roof Dead Load = $3.75 + 0.50 + 0.75 = 5.00\text{ kN/m}^2$

6.3 Live Load Assignment

- Office floors (Stories 1–4): 2.5 kN/m^2
- Ground floor lobby: 4.0 kN/m^2
- Roof (non-accessible area): 0.75 kN/m^2
- Corridors: 3.0 kN/m^2

6.4 Wind Load Analysis

Design Parameters:

- Basic Wind Speed (Vb): 47 m/s (Zone IV)
- Risk factor (k1): 1.0 | Terrain factor (k2): 0.97 | Topography (k3): 1.0
- Design Wind Speed: $V_d = k_1 \times k_2 \times k_3 \times V_b = 1.0 \times 0.97 \times 1.0 \times 47 = 45.6\text{ m/s}$
- Design Wind Pressure: $p_d = 0.6 \times V_d^2 / 1000 = 0.6 \times (45.6)^2 / 1000 = 1.25\text{ kN/m}^2$

Net Wind Force on Building = $p_d \times C_p \times A = 1.25 \times 1.0 \times (20 \times 17.5) = 437.5\text{ kN}$

6.5 Seismic Load Analysis

Base Shear Calculation (IS 1893:2016):

- Zone factor (Z): 0.16 | Importance factor (I): 1.0 | R: 5.0
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- Fundamental Time Period: $T = 0.075 \times h^{0.75} = 0.075 \times 17.5^{0.75} = 0.63 \text{ s}$
- $S_a/g = 1.36/T = 1.36/0.63 = 2.16$ (Medium Soil, 5% damping)
- Design Horizontal Coefficient: $A_h = (Z/2) \times (S_a/g) / (R/I) = (0.16/2) \times 2.16 / (5/1.0) = 0.035$

Seismic Weight (W) = $0.85 \times \text{Total Dead Load} = 0.85 \times 9,375 \text{ kN} = 7,969 \text{ kN}$

Base Shear (VB) = $A_h \times W = 0.035 \times 7,969 = 278.9 \text{ kN}$

7. Load Calculation Tables

7.1 Floor-wise Load Summary

Floor	DL (kN/m ²)	LL (kN/m ²)	Area (m ²)	Total DL (kN)	Total LL (kN)
Roof	5.00	0.75	300	1,500	225
4th	6.25	2.50	300	1,875	750
3rd	6.25	2.50	300	1,875	750
2nd	6.25	2.50	300	1,875	750
1st	6.25	2.50	300	1,875	750
G.F.	6.25	4.00	300	1,875	1,200
TOTAL	—	—	1,800	10,875	4,425

7.2 Factored Load Combinations (Governing)

Combination	Formula	Factored Load	Governs?
Gravity Only	1.4 DL	15,225 kN	No
Grav. + Live	1.2DL + 1.6LL	20,130 kN	YES ✓
Grav. + Wind	1.2DL + 1.0WL + LL	18,538 kN	No
Grav. + Seismic	1.2DL + 1.0EL + LL	17,827 kN	No
Uplift Check	0.9DL + 1.0WL	7,350 kN	No

7.3 Column Load Demand — Ground Floor (Critical)

Column ID	Trib. Area (m²)	Total DL (kN)	Total LL (kN)	Factored (kN)	FOS
C1 (Corner)	6.25	390	94	618	3.2
C2 (Edge)	12.50	780	188	1,237	3.1
C3 (Interior)	25.00	1,560	375	2,474	3.1
C4 (Edge)	12.50	780	188	1,237	3.1
C5 (Corner)	6.25	390	94	618	3.2

8. Failure Mode Analysis

8.1 Beam Failure Modes

Failure Mode	Cause	Detection Sign	Mitigation
Flexural Failure	Overloading in bending	Cracks at midspan soffit	Increase steel ratio, add depth
Shear Failure	Inadequate stirrups	Diagonal cracks near supports	Add shear reinforcement
Deflection Failure	Excessive span/depth	Visible sagging, cracking	Increase beam depth (L/20 rule)
Bond/Anchorage	Insufficient rebar development	Bar slippage, splitting cracks	Increase development length Ld

8.2 Column Failure Modes

- Short Column Failure — Crushing of concrete before steel yields: FOS ≥ 3.0 required.
- Long Column Buckling — Slenderness ratio (l_e/r) > 50 triggers buckling; use lateral bracing.
- Biaxial Bending — Corner columns subjected to bending in both axes; check with interaction diagram.
- Eccentric Loading — Offset axial loads create bending moments; must be accounted in design.

8.3 Slab Failure Modes

- Punching Shear — Critical around column heads; provide shear studs or drop panels.
 - Two-Way Flexure — Excessive span causing cracking in both directions; check L/d ratio ≤ 28.
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- Thermal Cracking — Temperature differential induces tension; provide minimum 0.12% temperature steel.

8.4 Overall Structural Failure Classification

Failure Type	Likelihood	Severity	Preventive Measure
Overloading	Low	High	Enforce design live load; restrict floor loading
Foundation Fail.	Low	Critical	FOS ≥ 3 on bearing; proper soil investigation
Wind Uplift	Medium	High	Anchor roof elements; check uplift combinations
Seismic Drift	Medium	High	Limit inter-story drift to $H/250$ (IS 1893)
Corrosion	Medium	Moderate	40 mm clear cover; use corrosion-resistant steel

9. Design Recommendations

9.1 Structural Member Sizing

- Columns: Minimum 400 mm $\times$ 400 mm for G+4 structures; use 450 mm $\times$ 450 mm for interior columns.
- Beams: Depth = Span/12 to Span/15 for continuous beams; breadth ≥ 230 mm.
- Slabs: 150 mm minimum for two-way slabs; check deflection with $L/d \leq 26$ (simply supported).
- Footings: Isolated footings sized for safe bearing pressure ≤ 150 kN/m² for medium soil.

9.2 Reinforcement Recommendations

- Minimum steel in beams: $0.85/f_y \times b d$ (IS 456 Cl. 26.5.1.1).
- Maximum steel in beams: $0.04 \times b d$ (4% of gross cross-section area).
- Column ties: spacing $\leq \min(300 \text{ mm, least lateral dimension, } 16 \times \text{bar dia})$.
- Development length: $L_d \geq (\phi \times \sigma_s) / (4 \times \tau_{bd})$ for tension bars.

9.3 Serviceability Checks

- Maximum beam deflection: $L/250$ under total load, $L/350$ after construction (IS 456 Cl. 23.2).
 - Crack width: Maximum 0.3 mm in moderate exposure; 0.2 mm in severe exposure.
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- Story drift: Maximum $H/400$ under wind; $H/250$ under seismic (IS 1893).

9.4 Durability Provisions

- Use M30 concrete minimum for moderate exposure (office building in urban area).
- Provide 40 mm clear cover to main reinforcement.
- Use corrosion-resistant steel (TMT Fe500D) in coastal/industrial environments.
- Apply epoxy coatings on exposed structural steel members.

9.5 Code Compliance Checklist

Requirement	Code Reference	Status
Dead & Live load per occupancy type	IS 875 Part 1 & 2	✓ Met
Wind load design & gust factor	IS 875 Part 3	✓ Met
Seismic zone & base shear calculation	IS 1893:2016	✓ Met
Concrete mix design & water-cement ratio	IS 456:2000	✓ Met
Minimum reinforcement requirements	IS 456 Cl. 26.5	✓ Met
Deflection limits ($L/250$, $L/350$)	IS 456 Cl. 23.2	✓ Met
Fire resistance rating (2-hour minimum)	IS 1641	✓ Met
Column slenderness ratio check	IS 456 Cl. 25.3	✓ Met

10. Conclusion

This Structural Analysis Project has successfully demonstrated the systematic methodology for studying loads and safety factors in a five-story reinforced concrete office building. The analysis covered all primary load types — dead, live, wind, and seismic — and applied both ASCE 7-22 and IS 456:2000/IS 875 load combination frameworks to derive factored design demands.

Key findings of this project include: the governing load combination for floor design is $1.2DL + 1.6LL$, yielding a factored total load of 20,130 kN; the seismic base shear of 278.9 kN is within the structural capacity of the SMRF system; and the minimum factor of safety of 3.1 for ground-floor columns exceeds the required minimum of 3.0, confirming structural adequacy.

The failure mode analysis identified punching shear at column-slab junctions and seismic inter-story drift as the two most critical design concerns. Both are addressed through the recommended reinforcement detailing and structural proportioning provided in Section 9.

This report reinforces the principle that safe structural design is not merely a matter of calculations, but of understanding the physical behavior of materials under load, recognizing potential failure mechanisms, and applying appropriate margins of safety that protect both lives and property throughout the service life of the structure.

11. References

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